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Source/Drain Device and Method of Forming Thereof

Abstract

A method of forming a semiconductor includes forming a first recess in a first semiconductor fin protruding from a substrate and forming a second recess in a second semiconductor fin protruding from the substrate first semiconductor fin and forming a source/drain region in the first recess. Forming the source/drain region includes forming a first portion of a first layer in the first recess and forming a second portion of the first layer in the second recess, forming a second layer on the first layer by flowing a first precursor, and forming a third layer on the second layer by flowing a second precursor, the third layer being a single continuous and the second recess material.

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Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE [0001] This application is a continuation of U.S. patent application Ser. No. 18/636,490, filed on Apr. 16, 2024, which is a divisional of U.S. patent application Ser. No. 17/458,950, filed on Aug. 27, 2021, now U.S. Pat. No. 11,990,511 issued May 21, 2024, each application is hereby incorporated herein by reference.

BACKGROUND

[0002] Semiconductor devices are used in a variety of electronic applications, such as, for example, personal computers, cell phones, digital cameras, and other electronic equipment. Semiconductor devices are typically fabricated by sequentially depositing insulating or dielectric layers, conductive layers, and semiconductor layers of material over a semiconductor substrate, and patterning the various material layers using lithography to form circuit components and elements thereon.

[0003] The semiconductor industry continues to improve the integration density of various electronic components (e.g., transistors, diodes, resistors, capacitors, etc.) by continual reductions in minimum feature size, which allow more components to be integrated into a given area.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0005] FIG. 1 illustrates an example of a FinFET in a three-dimensional view, in accordance with some embodiments.

[0006] FIGS. 2, 3A, 3B, 4, 5, 6A, 6B, 7, 8A, 8B, 9A, 9B, 9C, 10A, 10B, 10C, 11A, 11B, 11C, 12A, 12B, 12C, 13A, 13B, 13C, 13D, 13E, 13F, 14A, 14B, 14C, 15A, 15B, 15C, 16A, 16B, 16C, 17A, 17B, 17C, 17D, 18A, 18B, 18C, 19A, 19B, and 19C are cross-sectional views of intermediate stages in the manufacturing of FinFETs, in accordance with some embodiments.

DETAILED DESCRIPTION

[0007] The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0008] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and

the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated **90** degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0009] In accordance with some embodiments, increased merge heights of source/drain regions may be produced with epitaxial growth techniques including an intermediate source/drain layer. Larger air gaps underneath source/drain regions may be obtained, which may lead to reduced gate to source/drain capacitance and improved AC performance of subsequently formed devices. The epitaxial growth process including an intermediate source/drain layer may allow for a good process window, which may enable merging of neighboring source/drain regions without producing undesired electric shorts while also providing substantially flat top profiles of the source/drain regions to enable reduced contact resistance.

[0010] FIG. **1** illustrates an example of a FinFET in a three-dimensional view, in accordance with some embodiments. The FinFET comprises a fin **52** on a substrate **50** (e.g., a semiconductor substrate). Isolation regions **56** are disposed in the substrate **50**, and the fin **52** protrudes above and from between neighboring isolation regions **56**. Although the isolation regions **56** are described/illustrated as being separate from the substrate **50**, as used herein the term “substrate” may be used to refer to just the semiconductor substrate or a semiconductor substrate inclusive of isolation regions. Additionally, although the fin **52** is illustrated as a single, continuous material as the substrate **50**, the fin **52** and/or the substrate **50** may comprise a single material or a plurality of materials. In this context, the fin **52** refers to the portion extending between the neighboring isolation regions **56**.

[0011] A gate dielectric layer **92** is along sidewalls and over a top surface of the fin **52**, and a gate electrode **94** is over the gate dielectric layer **92**. Source/drain regions **82** are disposed in opposite sides of the fin **52** with respect to the gate dielectric layer **92** and gate electrode **94**. FIG. **1** further illustrates reference cross-sections that are used in later figures. Cross-section A-A is along a longitudinal axis of the gate electrode **94** and in a direction, for example, perpendicular to the direction of current flow between the source/drain regions **82** of the FinFET. Cross-section B-B is perpendicular to cross-section A-A and is along a longitudinal axis of the fin **52** and in a direction of, for example, a current flow between the source/drain regions **82** of the FinFET. Cross-section C-C is parallel to cross-section A-A and extends through a source/drain region of the FinFET. Subsequent figures refer to these reference cross-sections for clarity.

[0012] Some embodiments discussed herein are discussed in the context of FinFETs formed using a gate-last process. In other embodiments, a gate-first process may be used. Also, some embodiments contemplate aspects used in planar devices, such as planar FETs, nanostructure (e.g., nanosheet, nanowire, gate-all-around, or the like) field effect transistors (NSFETs), or the like.

[0013] FIGS. **2** through **19C** are cross-sectional views of intermediate stages in the manufacturing of FinFETs, in accordance with some embodiments. FIGS. **2** through **7** illustrate reference cross-section A-A illustrated in FIG. **1**, except for multiple fins/FinFETs. FIGS. **8A**, **9A**, **10A**, **11A**, **12A**, **13A**, **14A**, **15A**, **16A**, **17A**, **18A**, and **19A** are illustrated along reference cross-section A-A illustrated in FIG. **1**, and FIGS. **8B**, **9B**, **10B**, **11B**, **12B**, **13B**, **14B**, **14C**, **15B**, **16B**, **17B**, **17C**, **18B**, and **19B** are illustrated along a similar cross-section B-B illustrated in FIG. **1**, except for multiple fins/FinFETs. FIGS. **9C**, **10C**, **11C**, **12C**, **13C**, **13D**, **13E**, **13F**, **14C**, **15C**, **16C**, **17D**, **18C**, and **19C** are illustrated along reference cross-section C-C illustrated in FIG. **1**, except for multiple fins/FinFETs.

[0014] In FIG. **2**, a substrate **50** is provided. The substrate **50** may be a semiconductor substrate, such as a bulk semiconductor, a semiconductor-on-insulator (SOI) substrate, or the like, which may be doped (e.g., with a p-type or an n-type dopant) or undoped. The substrate **50** may be a wafer,

such as a silicon wafer. Generally, an SOI substrate is a layer of a semiconductor material formed on an insulator layer. The insulator layer may be, for example, a buried oxide (BOX) layer, a silicon oxide layer, or the like. The insulator layer is provided on a substrate, typically a silicon or glass substrate. Other substrates, such as a multi-layered or gradient substrate may also be used. In some embodiments, the semiconductor material of the substrate **50** may include silicon; germanium; a compound semiconductor including silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including silicon-germanium, gallium arsenide phosphide, aluminum indium arsenide, aluminum gallium arsenide, gallium indium arsenide, gallium indium phosphide, and/or gallium indium arsenide phosphide; or combinations thereof.

[0015] The substrate **50** has an n-type region **50N** and a p-type region **50P**. The n-type region **50N** can be for forming n-type devices, such as NMOS transistors, e.g., n-type FinFETs. The p-type region **50P** can be for forming p-type devices, such as PMOS transistors, e.g., p-type FinFETs. The n-type region **50N** may be physically separated from the p-type region **50P** (as illustrated by divider **21**), and any number of device features (e.g., other active devices, doped regions, isolation structures, etc.) may be disposed between the n-type region **50N** and the p-type region **50P**.

[0016] In FIG. 3A, fins **52** are formed in the substrate **50**. The fins **52** are semiconductor strips. In some embodiments, the fins **52** may be formed in the substrate **50** by etching trenches in the substrate **50**. The etching may be any acceptable etch process, such as a reactive ion etch (RIE), neutral beam etch (NBE), the like, or a combination thereof. The etch may be anisotropic.

[0017] The fins may be patterned by any suitable method. For example, the fins **52** may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the fins. In some embodiments, the mask (or other layer) may remain on the fins **52**.

[0018] FIG. 3B illustrates a region **55** with two single fins **52** and two double fins **53** extending from the substrate **50**, in accordance with some embodiments. The region **55** may be integrated in a same device as the regions **50N** and **50P** illustrated in and described with FIG. 3A, above. The region **55** may be used to form two double fin FinFETs and two single fin FinFETs. In some embodiments, the region **55** is used to form part of a static random access memory (SRAM) cell, such as an n-type SRAM cell. The single fins **52** and the double fins **53** may be formed with substantially similar materials and methods as the fins **52** described above with respect to FIG. 3A. The single fins **52** have a non-crown structure, in which a plurality of fins **52** are directly connected to the bulk portion of the substrate **50**. The double fins **53** have a crown structure comprising a semiconductor base **51** and fins **52** protruding from the semiconductor base. Although FIG. 3B illustrates two single fins **52** and two double fins **53**, any number of single fins **52** and double fins **53** may be formed.

[0019] In some embodiments, the double fins **53** are separated from each other by a width **W1** in a range of about 40 nm to about 50 nm, which may be advantageous to reduce formations of shorts by undesired merging of subsequently formed source/drain regions on adjacent double fins **53**.

[0020] In some embodiments, the protruding fins **52** of the double fins **53** are separated from each other by a width **W2** in a range of about 5 nm to about 30 nm. In some embodiments, the nearest single fin **52** and double fin **53** are separated by a width **W3** in a range of about 50 nm to about 200 nm. In some embodiments, adjacent single fins **52** are separated by a width **W4** in a range of about 15 nm to about 40 nm. Various subsequent figures and processing steps may be described with respect to any of Regions **50N**, **50P**, or **55**, and it should be understood that the same processing

steps are applied to each of the regions **50N**, **50P**, and **55** unless otherwise noted.

[0021] Following from FIG. **3A**, in FIG. **4**, an insulation material **54** is formed over the substrate **50** and between neighboring fins **52**. The insulation material **54** may be an oxide, such as silicon oxide, a nitride, the like, or a combination thereof, and may be formed by a high density plasma chemical vapor deposition (HDP-CVD), a flowable CVD (FCVD) (e.g., a CVD-based material deposition in a remote plasma system and post curing to make it convert to another material, such as an oxide), the like, or a combination thereof. Other insulation materials formed by any acceptable process may be used. In the illustrated embodiment, the insulation material **54** is silicon oxide formed by a FCVD process. An anneal process may be performed once the insulation material is formed. In an embodiment, the insulation material **54** is formed such that excess insulation material **54** covers the fins **52**. Although the insulation material **54** is illustrated as a single layer, some embodiments may utilize multiple layers. For example, in some embodiments a liner (not shown) may first be formed along a surface of the substrate **50** and the fins **52**. Thereafter, a fill material, such as those discussed above may be formed over the liner.

[0022] In FIG. **5**, a removal process is applied to the insulation material **54** to remove excess insulation material **54** over the fins **52**. In some embodiments, a planarization process such as a chemical mechanical polish (CMP), an etch-back process, combinations thereof, or the like may be utilized. The planarization process exposes the fins **52** such that top surfaces of the fins **52** and the insulation material **54** are level after the planarization process is complete. In embodiments in which a mask remains on the fins **52**, the planarization process may expose the mask or remove the mask such that top surfaces of the mask or the fins **52**, respectively, and the insulation material **54** are level after the planarization process is complete.

[0023] In FIG. **6A**, the insulation material **54** is recessed to form Shallow Trench Isolation (STI) regions **56**. The insulation material **54** is recessed such that upper portions of fins **52** in the n-type region **50N** and in the p-type region **50P** protrude from between neighboring STI regions **56**. Further, the top surfaces of the STI regions **56** may have a flat surface as illustrated, a convex surface, a concave surface (such as dishing), or a combination thereof. The top surfaces of the STI regions **56** may be formed flat, convex, and/or concave by an appropriate etch. The STI regions **56** may be recessed using an acceptable etching process, such as one that is selective to the material of the insulation material **54** (e.g., etches the material of the insulation material **54** at a faster rate than the material of the fins **52**). For example, an oxide removal using, for example, dilute hydrofluoric (dHF) acid may be used.

[0024] FIG. **6B** illustrates the region **55** after STI regions **56** are formed over the substrate and between neighboring single fins **52** and double fins **53**. The STI regions **56** may be formed over the semiconductor bases **51** and between the protruding fins **52** of the double fins **53**. In some embodiments as illustrated in FIG. **6B**, top surfaces of the STI regions **56** are flat between the protruding fins **52** of the double fins **53** and are concave between neighboring fins single fins **52** and between neighboring single fins **52** and double fins **53**. However, the top surfaces of the STI regions **56** may have flat surfaces, concave surfaces, convex surfaces, or a combination thereof.

[0025] The process described with respect to FIGS. **2** through **6B** is just one example of how the fins **52** may be formed. In some embodiments, the fins may be formed by an epitaxial growth process. For example, a dielectric layer can be formed over a top surface of the substrate **50**, and trenches can be etched through the dielectric layer to expose the underlying substrate **50**. Homoepitaxial structures can be epitaxially grown in the trenches, and the dielectric layer can be recessed such that the homoepitaxial structures protrude from the dielectric layer to form fins. Additionally, in some embodiments, heteroepitaxial structures can be used for the fins **52**. For example, the fins **52** in FIG. **5** can be recessed, and a material different from the fins **52** may be epitaxially grown over the recessed fins **52**. In such embodiments, the fins **52** comprise the recessed material as well as the epitaxially grown material disposed over the recessed material. In an even further embodiment, a dielectric layer can be formed over a top surface of the substrate **50**,

and trenches can be etched through the dielectric layer. Heteroepitaxial structures can then be epitaxially grown in the trenches using a material different from the substrate **50**, and the dielectric layer can be recessed such that the heteroepitaxial structures protrude from the dielectric layer to form the fins **52**. In some embodiments where homoepitaxial or heteroepitaxial structures are epitaxially grown, the epitaxially grown materials may be in situ doped during growth, which may obviate prior and subsequent implantations although in situ and implantation doping may be used together.

[0026] Still further, it may be advantageous to epitaxially grow a material in n-type region **50N** (e.g., an NMOS region) different from the material in p-type region **50P** (e.g., a PMOS region). In various embodiments, upper portions of the fins **52** may be formed from silicon-germanium ($\text{Si.sub.xGe.sub.1-x}$, where x can be in the range of 0 to 1), silicon carbide, pure or substantially pure germanium, a III-V compound semiconductor, a II-VI compound semiconductor, or the like. For example, the available materials for forming III-V compound semiconductor include, but are not limited to, indium arsenide, aluminum arsenide, gallium arsenide, indium phosphide, gallium nitride, indium gallium arsenide, indium aluminum arsenide, gallium antimonide, aluminum antimonide, aluminum phosphide, gallium phosphide, and the like.

[0027] Further in FIGS. **6A** and **6B**, appropriate wells (not shown) may be formed in the fins **52** and/or the substrate **50**. In some embodiments, a P well may be formed in the n-type region **50N**, and an N well may be formed in the p-type region **50P**. In some embodiments, a P well or an N well are formed in both the n-type region **50N** and the p-type region **50P**.

[0028] In the embodiments with different well types, the different implant steps for the n-type region **50N** and the p-type region **50P** may be achieved using a photoresist and/or other masks (not shown). For example, a photoresist may be formed over the fins **52** and the STI regions **56** in the n-type region **50N**. The photoresist is patterned to expose the p-type region **50P** of the substrate **50**. The photoresist can be formed by using a spin-on technique and can be patterned using acceptable photolithography techniques. Once the photoresist is patterned, an n-type impurity implant is performed in the p-type region **50P**, and the photoresist may act as a mask to substantially prevent n-type impurities from being implanted into the n-type region **50N**. The n-type impurities may be phosphorus, arsenic, antimony, or the like implanted in the region to a concentration of equal to or less than 10^{18} cm^{-3} , such as between about 10^{16} cm^{-3} and about 10^{18} cm^{-3} . After the implant, the photoresist is removed, such as by an acceptable ashing process.

[0029] Following the implanting of the p-type region **50P**, a photoresist is formed over the fins **52** and the STI regions **56** in the p-type region **50P**. The photoresist is patterned to expose the n-type region **50N** of the substrate **50**. The photoresist can be formed by using a spin-on technique and can be patterned using acceptable photolithography techniques. Once the photoresist is patterned, a p-type impurity implant may be performed in the n-type region **50N**, and the photoresist may act as a mask to substantially prevent p-type impurities from being implanted into the p-type region **50P**. The p-type impurities may be boron, boron fluoride, indium, or the like implanted in the region to a concentration of equal to or less than 10^{18} cm^{-3} , such as between about 10^{16} cm^{-3} and about 10^{18} cm^{-3} . After the implant, the photoresist may be removed, such as by an acceptable ashing process.

[0030] After the implants of the n-type region **50N** and the p-type region **50P**, an anneal may be performed to repair implant damage and to activate the p-type and/or n-type impurities that were implanted. In some embodiments, the grown materials of epitaxial fins may be in situ doped during growth, which may obviate the implantations, although in situ and implantation doping may be used together.

[0031] In FIG. **7**, a dummy dielectric layer **60** is formed on the fins **52**. The dummy dielectric layer **60** may be, for example, silicon oxide, silicon nitride, a combination thereof, or the like, and may be deposited or thermally grown according to acceptable techniques. A dummy gate layer **62** is formed over the dummy dielectric layer **60**, and a mask layer **64** is formed over the dummy gate

layer **62**. The dummy gate layer **62** may be deposited over the dummy dielectric layer **60** and then planarized, such as by a CMP. The mask layer **64** may be deposited over the dummy gate layer **62**. The dummy gate layer **62** may be a conductive or non-conductive material and may be selected from a group including amorphous silicon, polycrystalline-silicon (polysilicon), poly-crystalline silicon-germanium (poly-SiGe), metallic nitrides, metallic silicides, metallic oxides, and metals. The dummy gate layer **62** may be deposited by physical vapor deposition (PVD), CVD, sputter deposition, or other techniques for depositing the selected material. The dummy gate layer **62** may be made of other materials that have a high etching selectivity from the etching of isolation regions, e.g., the STI regions **56** and/or the dummy dielectric layer **60**. The mask layer **64** may include one or more layers of, for example, silicon nitride, silicon oxynitride, or the like. In this example, a single dummy gate layer **62** and a single mask layer **64** are formed across the n-type region **50N** and the p-type region **50P**. It is noted that the dummy dielectric layer **60** is shown covering only the fins **52** for illustrative purposes only. In some embodiments, the dummy dielectric layer **60** may be deposited such that the dummy dielectric layer **60** covers the STI regions **56**, extending over the STI regions and between the dummy gate layer **62** and the STI regions **56**.

[0032] FIGS. **8A** through **19C** illustrate various additional steps in the manufacturing of embodiment devices. FIGS. **8A** through **19C** illustrate features in either of the n-type region **50N** and the p-type region **50P**. For example, the structures illustrated in FIGS. **8A** through **19C** may be applicable to both the n-type region **50N** and the p-type region **50P**. Differences (if any) in the structures of the n-type region **50N** and the p-type region **50P** are described in the text accompanying each figure.

[0033] In FIGS. **8A** and **8B**, the mask layer **64** (see FIG. **7**) may be patterned using acceptable photolithography and etching techniques to form masks **74**. The pattern of the masks **74** then may be transferred to the dummy gate layer **62**. In some embodiments (not illustrated), the pattern of the masks **74** may also be transferred to the dummy dielectric layer **60** by an acceptable etching technique to form dummy gates **72**. The dummy gates **72** cover respective channel regions **58** of the fins **52**. The pattern of the masks **74** may be used to physically separate each of the dummy gates **72** from adjacent dummy gates. The dummy gates **72** may also have a lengthwise direction substantially perpendicular to the lengthwise direction of respective epitaxial fins **52**.

[0034] Further in FIGS. **8A** and **8B**, gate seal spacers **80** can be formed on exposed surfaces of the dummy gates **72**, the masks **74**, and/or the fins **52**. A thermal oxidation or a deposition followed by an anisotropic etch may form the gate seal spacers **80**. The gate seal spacers **80** may be formed of silicon oxide, silicon nitride, silicon oxynitride, or the like.

[0035] After the formation of the gate seal spacers **80**, implants for lightly doped source/drain (LDD) regions (not explicitly illustrated) may be performed. In the embodiments with different device types, similar to the implants discussed above in FIG. **6**, a mask, such as a photoresist, may be formed over the n-type region **50N**, while exposing the p-type region **50P**, and appropriate type (e.g., p-type) impurities may be implanted into the exposed fins **52** in the p-type region **50P**. The mask may then be removed. Subsequently, a mask, such as a photoresist, may be formed over the p-type region **50P** while exposing the n-type region **50N**, and appropriate type impurities (e.g., n-type) may be implanted into the exposed fins **52** in the n-type region **50N**. The mask may then be removed. The n-type impurities may be the any of the n-type impurities previously discussed, and the p-type impurities may be the any of the p-type impurities previously discussed. The lightly doped source/drain regions may have a concentration of impurities of from about 10^{15} cm.^{sup.}−3 to about 10^{19} cm.^{sup.}−3. An anneal may be used to repair implant damage and to activate the implanted impurities.

[0036] In FIGS. **9A**, **9B**, and **9C**, gate spacers **86** are formed on the gate seal spacers **80** along sidewalls of the dummy gates **72** and the masks **74**. The gate spacers **86** may be formed by conformally depositing an insulating material and subsequently anisotropically etching the insulating material. The insulating material of the gate spacers **86** may be silicon oxide, silicon

nitride, silicon oxynitride, silicon carbonitride, a combination thereof, or the like. In some embodiments as illustrated in FIG. 9C, the gate spacers **86** are further formed to cover a portion of the sidewalls of the fins **52** that extend above the STI regions **56**, which may be useful for guiding the subsequent epitaxial growth of source/drain regions (see below, FIGS. 10B-13D). In some other embodiments, the spacer etch used to form the gate spacers **86** may be adjusted to remove the spacer material to allow the subsequently epitaxially grown region to extend to the surface of the STI region **56**.

[0037] It is noted that the above disclosure generally describes a process of forming spacers and LDD regions. Other processes and sequences may be used. For example, fewer or additional spacers may be utilized, different sequence of steps may be utilized (e.g., the gate seal spacers **80** may not be etched prior to forming the gate spacers **86**, yielding “L-shaped” gate seal spacers, spacers may be formed and removed, and/or the like. Furthermore, the n-type and p-type devices may be formed using a different structures and steps. For example, LDD regions for n-type devices may be formed prior to forming the gate seal spacers **80** while the LDD regions for p-type devices may be formed after forming the gate seal spacers **80**.

[0038] In FIGS. 10A, 10B, and 10C recesses **81** are etched in the fins **52**. The etch process can be isotropic or anisotropic, and it may be selective with respect to one or more crystalline planes of the fin material. As a result, the recesses **81**, shown in FIGS. 10B and 10C as having round bottom profiles, can in practice have various profile shapes based on the etch process implemented. The etch process may be a dry etch, such as a RIE, NBE, or the like, and/or a wet etch, such as one using tetramethylammonium hydroxide (TMAH), ammonium hydroxide (NH₄OH), or other etchants. FIG. 10B illustrates the recesses **81** along cross-section B-B as illustrated in FIG. 1, and FIG. 10C illustrates the recesses **81** along cross-section C-C as illustrated in FIG. 1.

[0039] In FIGS. 11A through 17D, epitaxial source/drain regions **82** are formed in the fins **52**. In some embodiments, a material of the source/drain regions **82** is selected to exert stress in the respective channel regions **58**, thereby improving performance. The epitaxial source/drain regions **82** are formed in the fins **52** such that each dummy gate **72** is disposed between respective neighboring pairs of the epitaxial source/drain regions **82**. The epitaxial source/drain regions **82** may extend into the fins **52**. The gate spacers **86** may be used to separate the epitaxial source/drain regions **82** from the dummy gates **72** by an appropriate lateral distance so that the epitaxial source/drain regions **82** do not short out subsequently formed gates of the resulting FinFETs.

[0040] The epitaxial source/drain regions **82** in the region **50P**, e.g., the PMOS region, may be formed by masking the region **50N**, e.g., the NMOS region, and etching source/drain regions of the fins **52** in the region **50P** are etched to form recesses in the fins **52**. Then, the epitaxial source/drain regions **82** in the region **50P** are epitaxially grown in the recesses. The epitaxial source/drain regions **82** may include any acceptable material, such as appropriate for p-type FinFETs. For example, if the fin **52** is silicon, the epitaxial source/drain regions **82** in the region **50P** may comprise materials exerting a compressive strain in the channel region **58**, such as SiGe, SiGeB, Ge, GeSn, or the like. The epitaxial source/drain regions **82** in the region **50P** may also have surfaces raised from respective surfaces of the fins **52**. In an embodiment, the epitaxial source/drain regions **82** may be epitaxially grown in the region **50P** with a process that does not produce facets. In other embodiments, the epitaxial source/drain regions **82** may have facets.

[0041] The epitaxial source/drain regions **82** and/or the fins **52** may be implanted with dopants to form source/drain regions, similar to the process previously discussed for forming lightly-doped source/drain regions, followed by an anneal. The source/drain regions may have an impurity concentration of in a range of about 10^{19} cm⁻³ to about 10^{21} cm⁻³. The n-type and/or p-type impurities for source/drain regions may be any of the impurities previously discussed. In some embodiments, the epitaxial source/drain regions **82** may be in situ doped during growth.

[0042] FIGS. 11A through 13C illustrate the formation in the region **50N** of epitaxial source/drain

regions **82**. In some embodiments, each of the epitaxial source/drain regions **82** comprises a first layer **82A**, a second layer **82B**, and a third layer **82C** (as illustrated below in FIG. **13B**). In other embodiments, the epitaxial source/drain regions **82** may have a fewer or greater number of layers. [0043] FIGS. **11B** and **11C** illustrates the formation of the first layer **82A**, in accordance with some embodiments. Portions of the first layer **82A** may be formed on bottom surfaces of the recesses **81**. In some embodiments, the first layer **82A** comprises silicon doped with arsenic, silicon lightly doped with phosphorus, silicon carbide, silicon carbon phosphide, or the like. The dopant concentration of the As, P, C, or the C and P in the Si may be less than about 1.0×10^{21} atoms/cm³, such as in a range of about 1.0×10^{21} atoms/cm³ to about 6.0×10^{20} atoms/cm³, as measured with secondary ion mass spectrometry (SIMS). In some embodiments, the first layer **82A** has a dopant concentration of P less than 1.0×10^{21} atoms/cm³. The first layer **82A** may be epitaxially grown using metal-organic CVD (MOCVD), molecular beam epitaxy (MBE), liquid phase epitaxy (LPE), vapor phase epitaxy (VPE), selective epitaxial growth (SEG), a combination thereof, or the like. The first layer **82A** may be formed to a first thickness T1 in a range of about 2 nm to about 10 nm, measured as the shortest distance from a bottom point of the first layer **82A** to the top surface of the first layer **82A**. [0044] FIGS. **12B** and **12C** illustrate the formation of the second layer **82B**, in accordance with some embodiments. The second layer **82B** (also referred to as an intermediate source/drain layer) may be useful for raising the merge point of the portions of the epitaxial source/drain regions **82** (see below, FIG. **13C**) by increasing the epitaxial growth in the vertical direction. This may reduce the formation of disadvantageous electric shorts between adjacent epitaxial source/drain regions **82**, such as on neighboring double fins **53** (see below, FIG. **13E**).

[0045] The second layer **82B** may be epitaxially grown using metal-organic CVD (MOCVD), molecular beam epitaxy (MBE), liquid phase epitaxy (LPE), vapor phase epitaxy (VPE), selective epitaxial growth (SEG), a combination thereof, or the like. In some embodiments in which the second layer **82B** is formed of silicon doped by phosphorus, the epitaxial growth process uses a suitable Si precursor that promotes vertical growth of the second layer **82B** and suppresses lateral growth of the second layer **82B**. For example, the suitable Si precursor may be silane (SiH₄), DCS, the like, or a combination thereof. By using a silicon precursor that promotes vertical growth over lateral growth may be useful for producing bottom-up growth of the second layer **82B**, reducing epitaxial growth of the second layer **82B** on the sidewalls of the recess **81**, and reducing the lateral width of the subsequently formed third layer **82C** (see below, FIGS. **13B-C**), which may reduce the formation of disadvantageous electric shorts. The epitaxial growth process may also use a suitable P precursor to provide a phosphorus source for in situ doping the epitaxial source/drain regions **82** during the epitaxial growth process. The suitable P precursor may be phosphine, phosphorus oxychloride, the like, or a combination thereof. In some embodiments, carrier gases such as hydrogen (H₂) can be mixed with the precursors.

[0046] In some embodiments, the formation of the second layer **82B** uses a flow rate of the Si precursor in a range of about 20 sccm to about 50 sccm. A flow rate of the P precursor may be in a range of about 200 sccm to about 300 sccm. A flow rate of the carrier gas may be in a range of about 3000 sccm to about 10000 sccm. The epitaxial growth process may be performed at a temperature in a range of about 660° C. to about 720° C. and the epitaxial growth process may be performed at a pressure in a range of about 100 Torr to about 250 Torr. The formation process of the second layer **82B** may be performed for a duration in a range of about 10 seconds to about 50 seconds, which may be useful for selectivity.

[0047] The second layer **82B** may be formed to a second thickness T2 in a range of about 3 nm to about 7 nm, measured as the shortest distance from a bottom point of the second layer **82B** to the top surface of the second layer **82B**. The second thickness T2 being in a range of about 3 nm to about 7 nm may be advantageous for raising the merge point of the portions of the epitaxial source/drain regions **82** and reducing the formation of electrical shorts. The second thickness T2

being less than about 3 nm may be disadvantageous because the merge point of the portions of the epitaxial source/drain regions **82** may be insufficiently raised. The second thickness T2 being greater than about 7 nm may be disadvantageous because it may lead to increased selectivity loss. In some embodiments, the second layer **82B** is less than 0.5 nm thicker measured along a sidewall of the second layer **82B** than the thickness of the second layer measured as the shortest distance from a bottom point of the second layer **82B** to the top surface of the second layer **82B**.

[0048] The dopant concentration of phosphorus in the second layer **82B** may be in a range of about 3.0×10^{21} atoms/cm³ to about 3.5×10^{21} atoms/cm³, as measured with secondary ion mass spectrometry (SIMS), which may be advantageous because it may provide better selectivity. The dopant concentration of phosphorus in the second layer **82B** being less than about 3.0×10^{21} atoms/cm³ may be disadvantageous because it may lead to higher resistivity in the epitaxial source/drain regions **82**. The dopant concentration of phosphorus in the second layer **82B** being greater than about 3.5×10^{21} atoms/cm³ may be disadvantageous because it may lead to worse selectivity control. In some embodiments, the concentration of phosphorus in the second layer **82B** is greater than the concentration of phosphorus in the first layer **82A**.

[0049] In FIGS. **13B** and **13C**, a third layer **82C** is formed extending from the portions of the second layer **82B** and merges into a single continuous source/drain region **82** extending from two or more fins **52**, in accordance with some embodiments. As a result of the epitaxy processes used to form the third layer **82C**, upper surfaces of the third layer **82C** have facets which expand laterally outward beyond sidewalls of the fins **52**. These facets cause adjacent portions of the third layer **82C** of a same FinFET, such as FinFETs comprising double fins **53** as illustrated in FIG. **13E**, to merge as illustrated by FIG. **13C**. The third layer **82C** can be formed using an epitaxy process such as a cyclic deposition-etch (CDE) process. The CDE process may be useful for producing a source/drain region with a flatter top surface, which may decrease contact resistance and improve device performance. The CDE process includes a number of repeated cycles, such as in a range from 2 cycles to 5 cycles. Each cycle of the CDE process includes a deposition process followed by an etch process. In some embodiments, the deposition process of the CDE process includes a chemical vapor deposition (CVD) process, such as reduced pressure chemical vapor deposition (RPCVD), low pressure CVD (LPCVD), the like, or a combination thereof.

[0050] In some embodiments in which the third layer **82C** is formed of silicon doped by phosphorus, the epitaxial growth process uses a suitable Si precursor and a suitable P precursor. The P precursor provides a phosphorus source for in situ doping the epitaxial source/drain regions **82** during the epitaxial growth process. The suitable Si precursor may be selected to promote lateral growth of the third layer **82C** over vertical growth of the third layer **82C** such that the resulting third layer **82C** has a top surface with improved planarity. For example, the suitable precursor may be dichlorosilane (DCS), silane, the like, or a combination thereof. The suitable P precursor may be phosphine, phosphorus oxychloride, the like, or a combination thereof. In some embodiments, carrier gases such as hydrogen can be mixed with the precursors. Using a Si precursor that promotes lateral growth and hydrogen as the carrier gas may be useful for producing bottom up lateral growth of the third layer **82C** to merge the portions of the third layer **82C** grown on neighboring fins **52** and produce a flatter top profile of the source/drain regions **82**, which may decrease contact resistance and improve device performance. In some embodiments, vertical growth of the second layer **82B** is greater while flowing the suitable Si precursor for the second layer **82B** (such as e.g. silane) than vertical growth of the third layer **82C** while flowing the suitable Si precursor for the third layer **82C** (such as e.g. DCS), and lateral growth of the third layer **82C** is greater while flowing the suitable Si precursor for the third layer **82C** (such as e.g. DCS) than lateral growth of the second layer **82B** while flowing the suitable Si precursor for the second layer **82B** (such as e.g. silane).

[0051] In some embodiments, the formation of the third layer **82C** uses a flow rate of the Si

precursor in a range of about 800 sccm to about 1000 sccm. A flow rate of the dopant precursor may be in a range of about 200 sccm to about 300 sccm. A flow rate of the carrier gas may be in a range of about 3000 sccm to about 10000 sccm. The epitaxial growth process may be performed at a temperature in a range of about 660° C. to about 700° C. and the epitaxial growth process may be performed at a pressure in a range of about 200 Torr to about 350 Torr. The formation process of the third layer **82C** may be performed for a duration in a range of about 60 seconds to about 300 seconds, which may be useful for forming a layer of SiP with a sufficiently high P concentration. [0052] In some embodiments, the deposition process forms an epitaxial layer, such as e.g. SiP, on the second layer **82B** and an amorphous material on non-crystalline surfaces. In some embodiments, after the deposition process, a post-deposition purge operation is used to remove the deposition gases from the process chamber. An inert gas, such as He, Ar, or Ne, may be used in this operation to purge the deposition gases from the process chamber. Once the deposition gases are removed from the chamber, the etch process follows.

[0053] The etching (or partial etching) process of the CDE process removes the amorphous material and may also remove a portion of the deposited epitaxial layer. The remaining epitaxial layer forms the third layer **82C**. The etch process can be an in situ etch process performed in the chamber of the deposition process. In some embodiments, an etch gas is flowed into the chamber to etch the amorphous material. Etch gases such as HCl, GeH₄, the like, or a combination thereof can be used. A pressure during the etch process can be in a range from about 4 Torr to about 30 Torr. A temperature during the etch process can be in a range from about 720° C. to about 780° C. In some examples, after the etching process, a purge operation follows to remove the etching gases from the chamber.

[0054] The third layer **82C** may be formed to a fifth width W5 measured as a

[0055] maximum distance between opposite sidewalls in a range of about 53 to about 64 nm, such as about 57 to about 62 nm. The fifth width W5 being in a range of about 53 nm to about 64 nm may be advantageous for achieving a flatter top profile of the source/drain regions **82** that may decrease contact resistance and improve device performance. The fifth width W5 being less than about 53 nm may be disadvantageous because the top profile of the source/drain regions **82** may be insufficiently flat, leading to higher contact resistance. The fifth width W5 being greater than about **64** nm may be disadvantageous because it may lead to undesired merging between adjacent source/drain regions **82**, forming electrical shorts.

[0056] In some embodiments, the third layer **82C** merges to define an air gap **83** between the bottom surface of the third layer **82C** and the STI region **56**. The air gap **83** may have a first height H1 measured as the shortest distance from a top surface of the STI region **56** to a highest point (also referred to as a merge point) of the bottom surface of the third layer **82C** between neighboring fins **52**. The first height H1 may be in a range of about 30 nm to about 38 nm, which may be advantageous for reduced gate to source/drain capacitance, which may increase the AC performance of the subsequently formed device. For example, by increasing the first height H1 to the above range, an enlarged air gap **83** is defined, and parasitic capacitance can be reduced as a result of the air gap **83** having a low dielectric constant (e.g., approximately 1). In some embodiments the air gap **83** may include process gases used during the epitaxial growth of the third layer **82C**. For example, the air gap **83** may comprise DCS, phosphine, hydrogen, nitrogen or the like. The first height H1 being less than about 30 nm may be disadvantageous because the merge point of the portions of the epitaxial source/drain regions **82** may be insufficiently raised, which may lead to a smaller volume of air in the air gap **83**, increased gate to source/drain capacitance, and poorer AC performance of the subsequently formed device. The first height H1 being greater than about 32 nm may be disadvantageous it may lead to the epitaxially grown SiP layers being broken between neighboring fins **52**, which may cause high contact resistance in subsequently formed metal contacts. In some embodiments, the air gap **83** is subsequently filled by a first interlayer dielectric (ILD) **88** (see below, FIGS. **14A-14C**). In other embodiments, the air gap **83** is

not filled by the first ILD **88** and remains filled with air and/or remnants of process gases [0057] By forming the second layer **82B** (see above, FIGS. **12B-C**) with a silicon precursor such as e.g. silane that promotes vertical growth over lateral growth, the height of the third layer **82C** may be raised, increasing the height **H1** of the air gap **83** and raising the height of the merge point. This may be advantageous for reducing parasitic capacitance, which may increase the AC performance of the subsequently formed device.

[0058] In some embodiments, the top surface of the third layer **82C** has a valley between the fins **52**. The third layer **82C** may have a second height **H2** measured as the shortest distance from a lowest point of the valley to a surface level with the highest point of the third layer **82C** in a range of less than about 16 nm. A second height **H2** less than about 16 nm may be advantageous because it may provide a relatively flat landing site for a subsequently formed source/drain contact (see below, FIGS. **19A-C**), which may decrease contact resistance and improve device performance. A second height **H2** greater than about 16 nm may be disadvantageous because it may provide an uneven landing site for a subsequently formed source/drain contact, which may increase contact resistance and reduce device performance.

[0059] A third height **H3**, also referred to as a merge thickness, is measured as the shortest distance between the merge point of the bottom surface of the third layer **82C** and the top surface of the third layer **82C**. The third height **H3** may be in a range of about 20 nm to about 35 nm, which may be advantageous for reducing gate to source/drain capacitance and improving device performance. A third height **H3** less than about 20 nm may be disadvantageous because it may increase the likelihood of the source/drain region being damaged by a subsequent etching in order to form a metal contact etching (see below, FIGS. **19A-C**), which may induce high resistivity. A third height **H3** greater than about 35 nm may be disadvantageous because it may increase the height of the source/drain regions beyond desired dimensions.

[0060] The dopant concentration of the dopant, such as e.g. phosphorus, in the third layer **82C** may be in a range of about 2.5×10^{21} atoms/cm³ to about 4.0×10^{21} atoms/cm³, as measured with secondary ion mass spectrometry (SIMS), which may be advantageous because it may have low resistivity. The dopant concentration of phosphorus in the third layer **82C** being less than about 2.5×10^{21} atoms/cm³ may be disadvantageous because it may result in high resistivity. The dopant concentration of phosphorus in the third layer **82C** being greater than about 4.0×10^{21} atoms/cm³ may be disadvantageous because it may have a poor selectivity window with respect to subsequently formed layers adjacent to the third layer **82C**. In some embodiments, the concentration of phosphorus in the third layer **82C** is greater than the concentration of phosphorus in the first layer **82A**, and the concentration of phosphorus in the third layer **82C** may also be greater than the concentration of phosphorus in the second layer **82B**.

[0061] In some embodiments, additional layers (not illustrated) are formed over the third layer **82C** to form the epitaxial source/drain regions **82**. The additional layers may be doped silicon layers that substantially similar to the first layer **82A**, the second layer **82B**, or the third layer **82C** and/or capping or protection layers comprising silicon. The additional layers may be formed using substantially similar methods and materials as the first layer **82A**, the second layer **82B**, or the third layer **82C**.

[0062] FIG. **13D** illustrates epitaxial source/drain regions **82** formed extending from single fins **52** that do not merge with neighboring source/drain regions **82** on neighboring fins, in accordance with some embodiments. The source/drain regions **82** may be formed with substantially similar processes and materials as described above for the source/drain regions **82** with respect to FIGS. **11B** through **13C**, but the source/drain regions **82** do not merge with each other due to the single fins **52** being separated by a greater distance than the fins **52** illustrated in FIG. **13C** above.

[0063] FIG. **13E** illustrates the region **55** after epitaxial source/drain regions **82** are formed extending from the single fins **52** and double fins **53**, in accordance with some embodiments. The source/drain regions **82** extending from the double fins **53** may merge as illustrated above in FIG.

13C due to the protruding fins **52** being separated by a width **W2** in a range of about 12 nm to about 25 nm. The source/drain regions **82** extending from the single fins **52** may not merge due to the single fins **52** being separated by a width **W4** in a range of about 30 nm to about 40 nm. In some embodiments, the single fins **52** with separate source/drain regions **82** and the double fins **53** with merged source/drain regions **82** may be used to form part of an SRAM cell, such as an n-type SRAM cell.

[0064] FIG. **13F** illustrates a merged epitaxial source/drain region **82** extending from and merged across three neighboring fins **52**, in accordance with some embodiments. The merged epitaxial source/drain region **82** of FIG. **13F** may be formed using substantially similar methods and materials as described for the source/drain region **82** with respect to FIGS. **11B** to **13C**, except for being merged across three fins **52** due to the neighboring fins **52** being separated by widths **W2** in a range of about 12 nm to about 25 nm. In some embodiments, a merged source/drain region **82** may be formed extending from four or more neighboring fins **52**.

[0065] In FIGS. **14A**, **14B**, and **14C**, a first interlayer dielectric (ILD) **88** is deposited over the structure illustrated in FIGS. **13A**, **13B**, and **13C**. The first ILD **88** may be formed of a dielectric material, and may be deposited by any suitable method, such as CVD, plasma-enhanced CVD (PECVD), or FCVD. Dielectric materials may include phospho-silicate glass (PSG), boro-silicate glass (BSG), boron-doped phospho-silicate glass (BPSG), undoped silicate glass (USG), or the like. Other insulation materials formed by any acceptable process may be used. In some embodiments, a contact etch stop layer (CESL) **87** is disposed between the first ILD **88** and the epitaxial source/drain regions **82**, the masks **74**, and the gate spacers **86**. The CESL **87** may comprise a dielectric material, such as, silicon nitride, silicon oxide, silicon oxynitride, or the like, having a lower etch rate than the material of the overlying first ILD **88**. In some embodiments, the CESL **87** and the first ILD **88** fill the air gap **83**, as illustrated in FIG. **14C**. In other embodiments (not illustrated), the CESL **87** and/or the first ILD **88** do not fill the air gap **83**, and the air gap **83** remains filled with air and/or remnants of process gases.

[0066] In FIGS. **15A**, **15B**, and **15C**, a planarization process, such as a CMP, may be performed to level the top surface of the first ILD **88** with the top surfaces of the dummy gates **72** or the masks **74**. The planarization process may also remove the masks **74** on the dummy gates **72**, and portions of the gate seal spacers **80** and the gate spacers **86** along sidewalls of the masks **74**. After the planarization process, top surfaces of the dummy gates **72**, the gate seal spacers **80**, the gate spacers **86**, and the first ILD **88** are level. Accordingly, the top surfaces of the dummy gates **72** are exposed through the first ILD **88**. In some embodiments, the masks **74** may remain, in which case the planarization process levels the top surface of the first ILD **88** with the top surfaces of the top surface of the masks **74**.

[0067] In FIGS. **16A**, **16B**, and **16C**, the dummy gates **72**, and the masks **74** if present, are removed in an etching step(s), so that recesses **90** are formed. Portions of the dummy dielectric layer **60** in the recesses **90** may also be removed. In some embodiments, only the dummy gates **72** are removed and the dummy dielectric layer **60** remains and is exposed by the recesses **90**. In some embodiments, the dummy dielectric layer **60** is removed from recesses **90** in a first region of a die (e.g., a core logic region) and remains in recesses **90** in a second region of the die (e.g., an input/output region). In some embodiments, the dummy gates **72** are removed by an anisotropic dry etch process. For example, the etching process may include a dry etch process using reaction gas(es) that selectively etch the dummy gates **72** with little or no etching of the first ILD **88** or the gate spacers **86**. Each recess **90** exposes and/or overlies a channel region **58** of a respective fin **52**. Each channel region **58** is disposed between neighboring pairs of the epitaxial source/drain regions **82**. During the removal, the dummy dielectric layer **60** may be used as an etch stop layer when the dummy gates **72** are etched. The dummy dielectric layer **60** may then be optionally removed after the removal of the dummy gates **72**.

[0068] In FIGS. **17A** and **17B**, gate dielectric layers **92** and gate electrodes **94** are formed for

replacement gates. FIG. 17C illustrates a detailed view of region 89 of FIG. 17B. FIG. 17D illustrates a view of the source/drain regions 82 in cross-section C-C. Gate dielectric layers 92 one or more layers deposited in the recesses 90, such as on the top surfaces and the sidewalls of the fins 52 and on sidewalls of the gate seal spacers 80/gate spacers 86. The gate dielectric layers 92 may also be formed on the top surface of the first ILD 88. In some embodiments, the gate dielectric layers 92 comprise one or more dielectric layers, such as one or more layers of silicon oxide, silicon nitride, metal oxide, metal silicate, or the like. For example, in some embodiments, the gate dielectric layers 92 include an interfacial layer of silicon oxide formed by thermal or chemical oxidation and an overlying high-k dielectric material, such as a metal oxide or a silicate of hafnium, aluminum, zirconium, lanthanum, manganese, barium, titanium, lead, and combinations thereof. The gate dielectric layers 92 may include a dielectric layer having a k value greater than about 7.0. The formation methods of the gate dielectric layers 92 may include Molecular-Beam Deposition (MBD), ALD, PECVD, and the like. In embodiments where portions of the dummy gate dielectric 60 remains in the recesses 90, the gate dielectric layers 92 include a material of the dummy gate dielectric 60 (e.g., SiO₂).

[0069] The gate electrodes 94 are deposited over the gate dielectric layers 92, respectively, and fill the remaining portions of the recesses 90. The gate electrodes 94 may include a metal-containing material such as titanium nitride, titanium oxide, tantalum nitride, tantalum carbide, cobalt, ruthenium, aluminum, tungsten, combinations thereof, or multi-layers thereof. For example, although a single layer gate electrode 94 is illustrated in FIG. 17B, the gate electrode 94 may comprise any number of liner layers 94A, any number of work function tuning layers 94B, and a fill material 94C as illustrated by FIG. 17C. After the filling of the recesses 90, a planarization process, such as a CMP, may be performed to remove the excess portions of the gate dielectric layers 92 and the material of the gate electrodes 94, which excess portions are over the top surface of the ILD 88. The remaining portions of material of the gate electrodes 94 and the gate dielectric layers 92 thus form replacement gates of the resulting FinFETs. The gate electrodes 94 and the gate dielectric layers 92 may be collectively referred to as a “gate stack.” The gate and the gate stacks may extend along sidewalls of a channel region 58 of the fins 52.

[0070] The formation of the gate dielectric layers 92 in the n-type region 50N and the p-type region 50P may occur simultaneously such that the gate dielectric layers 92 in each region are formed from the same materials, and the formation of the gate electrodes 94 may occur simultaneously such that the gate electrodes 94 in each region are formed from the same materials. In some embodiments, the gate dielectric layers 92 in each region may be formed by distinct processes, such that the gate dielectric layers 92 may be different materials, and/or the gate electrodes 94 in each region may be formed by distinct processes, such that the gate electrodes 94 may be different materials. Various masking steps may be used to mask and expose appropriate regions when using distinct processes.

[0071] In FIGS. 18A, 18B, and 18C, a gate mask 96 is formed over the gate stack (including a gate dielectric layer 92 and a corresponding gate electrode 94), and the gate mask may be disposed between opposing portions of the gate spacers 86. In some embodiments, forming the gate mask 96 includes recessing the gate stack so that a recess is formed directly over the gate stack and between opposing portions of gate spacers 86. A gate mask 96 comprising one or more layers of dielectric material, such as silicon nitride, silicon oxynitride, or the like, is filled in the recess, followed by a planarization process to remove excess portions of the dielectric material extending over the first ILD 88.

[0072] As also illustrated in FIGS. 18A, 18B, and 18C, a second ILD 108 is deposited over the first ILD 88. In some embodiments, the second ILD 108 is a flowable film formed by a flowable CVD method. In some embodiments, the second ILD 108 is formed of a dielectric material such as PSG, BSG, BPSG, USG, or the like, and may be deposited by any suitable method, such as CVD and PECVD. The subsequently formed gate contacts 110 (FIGS. 16A and 16B) penetrate through the

second ILD **108** and the gate mask **96** to contact the top surface of the recessed gate electrode **94**. [0073] In FIGS. **19A**, **19B**, and **19C**, gate contacts **110** and source/drain contacts **112** are formed through the second ILD **108** and the first ILD **88** in accordance with some embodiments. Openings for the source/drain contacts **112** are formed through the first and second ILDs **88** and **108**, and openings for the gate contact **110** are formed through the second ILD **108** and the gate mask **96**. The openings may be formed using acceptable photolithography and etching techniques. A liner (not shown), such as a diffusion barrier layer, an adhesion layer, or the like, and a conductive material are formed in the openings. The liner may include titanium, titanium nitride, tantalum, tantalum nitride, or the like. The conductive material may be copper, a copper alloy, silver, gold, tungsten, cobalt, aluminum, nickel, or the like. A planarization process, such as a CMP, may be performed to remove excess material from a surface of the ILD **108**. The remaining liner and conductive material form the source/drain contacts **112** and gate contacts **110** in the openings. An anneal process may be performed to form a silicide **111** at the interface between the epitaxial source/drain regions **82** and the source/drain contacts **112**. The source/drain contacts **112** are physically and electrically coupled to the epitaxial source/drain regions **82**, and the gate contacts **110** are physically and electrically coupled to the gate electrodes **106**. The source/drain contacts **112** and gate contacts **110** may be formed in different processes, or may be formed in the same process. Although shown as being formed in the same cross-sections, it should be appreciated that each of the source/drain contacts **112** and gate contacts **110** may be formed in different cross-sections, which may avoid shorting of the contacts.

[0074] The disclosed FinFET embodiments could also be applied to nanostructure devices such as nanostructure (e.g., nanosheet, nanowire, gate-all-around, or the like) field effect transistors (NSFETs). In an NSFET embodiment, the fins are replaced by nanostructures formed by patterning a stack of alternating layers of channel layers and sacrificial layers. Dummy gate stacks and source/drain regions are formed in a manner similar to the above-described embodiments. After the dummy gate stacks are removed, the sacrificial layers can be partially or fully removed in channel regions. The replacement gate structures are formed in a manner similar to the above-described embodiments, the replacement gate structures may partially or completely fill openings left by removing the sacrificial layers, and the replacement gate structures may partially or completely surround the channel layers in the channel regions of the NSFET devices. ILDs and contacts to the replacement gate structures and the source/drain regions may be formed in a manner similar to the above-described embodiments. A nanostructure device can be formed as disclosed in U.S. Patent Application Publication No. 2016/0365414, which is incorporated herein by reference in its entirety.

[0075] Embodiments may provide advantages. Large air gaps underneath source/drain regions may be produced with epitaxial growth techniques including an intermediate source/drain layer. Resulting increased merge heights of the source/drain regions may lead to reduced MOSFET gate to source/drain parasitic capacitance and improved AC performance of subsequently formed devices. The method of production may allow for a good process window, which may enable merging of neighboring source/drain regions without producing undesired electric shorts while also producing substantially flat top profiles of the source/drain regions to enable reduced contact resistance.

[0076] In accordance with an embodiment, a method of forming a semiconductor device, the method includes: forming a first gate structure over a first semiconductor fin and a second semiconductor fin, the first semiconductor fin and the second semiconductor fin protruding from a substrate; forming a first recess in the first semiconductor fin adjacent the first gate structure and forming a second recess in the second semiconductor fin adjacent the first gate structure; and forming a first source/drain region in the first recess and the second recess. The forming the first source/drain region includes: forming a first layer in the first recess and the second recess, the forming the first layer including forming a first portion of the first layer in the first recess and

forming a second portion of the first layer in the second recess; forming a second layer on the first layer by flowing a silane gas, the forming the second layer including forming a third portion of the second layer on the first portion of the first layer and forming a fourth portion of the second layer on the second portion of the first layer; and forming a third layer on the second layer, the forming the third layer including flowing a dichlorosilane gas, the third layer being a single continuous material extending from the third portion of the second layer to the fourth portion of the second layer. In an embodiment, the first layer is formed to a thickness in a range of 2 nm to 10 nm. In an embodiment, the first layer includes Si doped with a dopant, the dopant including P, As, or C. In an embodiment, forming the second layer further includes flowing a phosphorus precursor, the phosphorus precursor including phosphine or phosphorus oxychloride. In an embodiment, the second layer has a thickness in a range of 3 nm to 7 nm. In an embodiment, forming the third layer includes a cyclic deposition-etch (CDE) process. In an embodiment, forming the third layer includes forming an air gap under the third layer. In an embodiment, a height of the air gap is in a range of 30 nm to 38 nm. In an embodiment, the method further includes the first source/drain region being formed in a third recess in a third semiconductor fin, the third semiconductor fin being adjacent the second semiconductor fin, a fifth portion of the first layer of the first source/drain region being formed in the third recess, a sixth portion of the second layer of the first source/drain region being formed on the fifth portion of the first layer, and the third layer physically contacting the sixth portion of the second layer.

[0077] In accordance with another embodiment, a method of forming a semiconductor device includes: forming a first recess in a first semiconductor fin, the first semiconductor fin protruding from a substrate; forming a second recess in a second semiconductor fin, the second semiconductor fin protruding from the substrate adjacent the first semiconductor fin; forming a first portion of a first source/drain layer in the first recess and forming a second portion of the first source/drain layer in the second recess; forming a first portion of a second source/drain layer on the first portion of the first source/drain layer and forming a second portion of the second source/drain layer on the second portion of the first source/drain layer, wherein forming the second source/drain layer includes flowing a first precursor with a hydrogen ambient; forming a third source/drain layer on the first portion of the second source/drain layer and the second portion of the second source/drain layer, the third source/drain layer being a single continuous layer extending from the first recess to the second recess, wherein forming the third layer including flowing a second precursor, wherein vertical growth of the second source/drain layer is greater while flowing the first precursor than vertical growth of the third layer while flowing the second precursor, and wherein lateral growth of the third source/drain layer is greater while flowing the second precursor than lateral growth of the second source/drain layer while flowing the first precursor; and forming a gate structure over the first semiconductor fin and the second semiconductor fin. In an embodiment, the first precursor is silane and the second precursor is dichlorosilane. In an embodiment, the second source/drain layer is doped with phosphorus to a density of phosphorus in a range of 3.0×10^{21} atoms/cm³ to 3.5×10^{21} atoms/cm³. In an embodiment, the second source/drain layer has a first thickness measured as the shortest distance from a bottom point of the second source/drain layer to the top surface of the second source/drain layer, the second source/drain layer has a second thickness measured along a sidewall of the second source/drain layer, and the second thickness is less than 0.5 nm greater than the first thickness. In an embodiment, the third layer has a first width in a range of 53 to 64 nm. In an embodiment, the top surface of the third source/drain layer has a valley between the first semiconductor fin and the second semiconductor fin, the third source/drain layer has a first height measured as the shortest distance from a lowest point of the valley to a surface level with the highest point of the third source/drain layer, and the first height is less than 16 nm. In an embodiment, the third source/drain layer has a second height measured as the shortest distance between a merge point of the third source/drain layer and the bottom surface of the third source/drain layer, the merge point being the highest point of the bottom surface of the third

source/drain layer between the first semiconductor fin and the second semiconductor fin, the second height being in a range of 20 nm to 35 nm. In an embodiment, the third source/drain layer is doped with phosphorus to a density of phosphorus in a range of 2.5×10^{21} atoms/cm³ to 4.0×10^{21} atoms/cm³.

[0078] In accordance with yet another embodiment, a semiconductor device includes: a first semiconductor fin and a second semiconductor fin, the first semiconductor fin and the second semiconductor fin extending from a substrate; an isolation region disposed on the substrate between the first semiconductor fin and the second semiconductor fin; a gate electrode over the first semiconductor fin and the second semiconductor fin; and a source/drain region adjacent to the gate electrode and over the first semiconductor fin and the second semiconductor fin. The source/drain region includes: a first layer, a first portion of the first layer extending from the first semiconductor fin, a second portion of the first layer extending from the second semiconductor fin; a second layer, a third portion of the second layer extending from the first portion of the first layer, a fourth portion of the second layer extending from the second portion of the first layer; and a third layer, the third layer being a single continuous material extending from the third portion of the second layer to the fourth portion of the second layer, the third layer having a first height greater than 30 nm, the first height being measured as the shortest distance from a top surface of the isolation region to a merge point of the third layer, the merge point being a highest point of the bottom surface of the third layer between the first semiconductor fin and the second semiconductor fin. In an embodiment, the first layer has a density of phosphorus less than 1×10^{21} atoms/cm³, the second layer has a density of phosphorus in a range of 3.0×10^{21} atoms/cm³ to 3.5×10^{21} atoms/cm³, and the third layer has a density of phosphorus in a range of 2.5×10^{21} atoms/cm³ to 4.0×10^{21} atoms/cm³. In an embodiment, the semiconductor device further includes an air gap between the source/drain region and the isolation region, the air gap having a height in a range of 30 nm to 38 nm.

[0079] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor device, comprising: a first semiconductor fin and a second semiconductor fin, wherein the first semiconductor fin and the second semiconductor fin extend from a substrate; an isolation region disposed on the substrate, wherein the isolation region is between the first semiconductor fin and the second semiconductor fin; and a source/drain region on the first semiconductor fin and the second semiconductor fin, the source/drain region comprising: a first layer, wherein a first portion of the first layer is on the first semiconductor fin, wherein a second portion of the first layer is on the second semiconductor fin, and wherein the first portion of the first layer is separated from the second portion of the first layer; a second layer, wherein a first portion of the second layer is on the first portion of the first layer, wherein a second portion of the second layer is on the second portion of the first layer, and wherein the first portion of the second layer is separated from the second portion of the second layer; and a third layer continuously extending from the first portion of the second layer to the second portion of the second layer.
2. The semiconductor device of claim 1, wherein the third layer is separated from the isolation region by a dielectric layer.

3. The semiconductor device of claim 1, wherein the third layer is separated from the isolation region by an air gap.
4. The semiconductor device of claim 1, wherein the first layer, the second layer, and the third layer comprise phosphorus.
5. The semiconductor device of claim 4, wherein the first layer has a first concentration of phosphorus and the second layer has a second concentration of phosphorus, and wherein the second concentration of phosphorus is greater than the first concentration of phosphorus.
6. The semiconductor device of claim 5, wherein the third layer has a third concentration of phosphorus, and wherein the third concentration of phosphorus is greater than the second concentration of phosphorus.
7. The semiconductor device of claim 1, wherein the third layer has a concave upper surface, wherein a vertical distance between a highest point of the concave upper surface of the third layer and a lowest point of the concave upper surface of the third layer is less than 16 nm.
8. A semiconductor device, comprising: a first semiconductor fin and a second semiconductor fin; an isolation region disposed between the first semiconductor fin and the second semiconductor fin; a source/drain region on the first semiconductor fin and the second semiconductor fin, the source/drain region comprising: a first layer, wherein a first portion of the first layer is on the first semiconductor fin and a second portion of the first layer is on the second semiconductor fin; a second layer, wherein a first portion of the second layer is on the first portion of the first layer and a second portion of the second layer is on the second portion of the first layer; and a third layer, wherein the third layer connects the first portion of the second layer and the second portion of the second layer; and a spacer along a sidewall of the first portion of the first layer, a sidewall of the first portion of the second layer, and a sidewall of the third layer.
9. The semiconductor device of claim 8, wherein the first portion of the first layer is separated from the second portion of the first layer, and wherein the first portion of the second layer is separated from the second portion of the second layer.
10. The semiconductor device of claim 8, wherein the spacer covers an interface between the first portion of the first layer and the first portion of the second layer, and wherein the spacer covers an interface between the first portion of the second layer and the third layer.
11. The semiconductor device of claim 8, wherein the third layer has a concave lower surface and a concave upper surface.
12. The semiconductor device of claim 11, wherein the third layer has a first height in a range from 30 nm to 38 nm, wherein the first height is measured as a shortest distance from a top surface of the isolation region to a highest point of the concave lower surface.
13. The semiconductor device of claim 8, wherein a concentration of phosphorus in the second layer is larger than a concentration of phosphorus in the first layer, and wherein a concentration of phosphorus in the third layer is larger than the concentration of phosphorus in the second layer.
14. A semiconductor device, comprising: a first semiconductor fin and a second semiconductor fin; and a source/drain feature on the first semiconductor fin and the second semiconductor fin, the source/drain feature comprising: a first layer, wherein a first portion of the first layer is in contact with the first semiconductor fin and a second portion of the first layer is in contact with the second semiconductor fin; a second layer, wherein a first portion of the second layer is in contact with the first portion of the first layer and a second portion of the second layer is in contact with the second portion of the first layer; and a third layer, wherein the third layer is a single continuous layer in contact with the first portion of the second layer and the second portion of the second layer, wherein the third layer has a curved upper surface and a curved lower surface in a cross-sectional view; an interlayer dielectric layer over the source/drain feature; a source/drain contact extending through the interlayer dielectric layer to electrically couple to the source/drain feature; and a metal silicide layer between the source/drain feature and the source/drain contact, wherein the metal silicide layer comprises a curved profile in the cross-sectional view.

- 15.** The semiconductor device of claim 14, further comprising an isolation region disposed between the first semiconductor fin and the second semiconductor fin, wherein an air gap is between a lower surface of the third layer and the isolation region.
- 16.** The semiconductor device of claim 14, further comprising a spacer in contact with a sidewall of the first portion of the first layer, a sidewall of the first portion of the second layer, and a sidewall of the third layer, wherein a top portion of the spacer is above a top surface of the first portion of the second layer, and wherein a bottom portion of the spacer is below a bottom surface of the first portion of the second layer.
- 17.** The semiconductor device of claim 14, wherein the first portion of the second layer is separated from the second portion of the second layer.
- 18.** The semiconductor device of claim 14, wherein the first layer has a thickness in a range from 2 nm to 10 nm.
- 19.** The semiconductor device of claim 14, wherein the second layer has a thickness in a range from 3 nm to 7 nm.
- 20.** The semiconductor device of claim 19, wherein a vertical distance between a highest point of the curved lower surface of the third layer and a lowest point of the curved upper surface of the third layer is in a range from 20 nm to 35 nm.
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